QUANTUM ARCHITECT · Electron Configuration Practice · ANSWER KEY — TEACHER COPY

Instructions: Answer each item in the space provided. Use the notation discussed in class. Show your work for any orbital diagram.

Teacher: _____	Class: _____	Elements: 12	Notation: shell	Total marks: 60

#	Question	Answer	Teacher note	Marks
1	Write the full electron configuration of Arsenic, As (Z = 33).	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^3$ shorthand: [Ar] $3d^{10}\ 4s^2\ 4p^3$	—	5
2	Write the noble-gas (shorthand) electron configuration of Selenium, Se (Z = 34).	[Ar] $3d^{10}\ 4s^2\ 4p^4$	Full form: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^{10}\ 4s^2\ 4p^4$. Core is Ar (Z = 18).	5
3	Draw the orbital diagram (boxes with ↑ and ↓ arrows) for Cobalt, Co (Z = 27).	1s [↑↓] 2s [↑↓] 2p [↑↓] [↑↓] [↑↓] 3s [↑↓] 3p [↑↓] [↑↓] [↑↓] 3d [↑↓] [↑↓] [↑] [↑] [↑] 4s [↑↓]	Hund's rule: single occupancy across the degenerate boxes before any pair. Pauli: a pair is always ↑↓.	5
4	Identify the element described by this configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$ Give its symbol, name and atomic number.	K, Potassium, Z = 19	Shorthand: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^1$. Shells: 2, 8, 8, 1.	5

#	Question	Answer	Teacher note	Marks
5	Write the full electron configuration of Manganese, Mn (Z = 25).	$1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^5\ 4s^2$ shorthand: [Ar] $3d^5\ 4s^2$	—	5
6	Write the noble-gas (shorthand) electron configuration of Vanadium, V (Z = 23).	[Ar] $3d^3\ 4s^2$	Full form: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 3d^3\ 4s^2$. Core is Ar (Z = 18).	5
7	Draw the orbital diagram (boxes with ↑ and ↓ arrows) for Sodium, Na (Z = 11).	1s [↑↓] 2s [↑↓] 2p [↑↓] [↑↓] [↑↓] 3s [↑]	Hund's rule: single occupancy across the degenerate boxes before any pair. Pauli: a pair is always ↑↓.	5
8	Identify the element described by this configuration: $1s^1$ Give its symbol, name and atomic number.	H, Hydrogen, Z = 1	Shorthand: $1s^1$. Shells: 1.	5

#	Question	Answer	Teacher note	Marks
9	Write the full electron configuration of Neon, Ne (Z = 10).	$1s^2\ 2s^2\ 2p^6$ shorthand: [He] $2s^2\ 2p^6$	—	5
10	Write the noble-gas (shorthand) electron configuration of Sulfur, S (Z = 16).	[Ne] $3s^2\ 3p^4$	Full form: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^4$. Core is Ne (Z = 10).	5
11	Draw the orbital diagram (boxes with ↑ and ↓ arrows) for Lithium, Li (Z = 3).	1s [↑↓] 2s [↑]	Hund's rule: single occupancy across the degenerate boxes before any pair. Pauli: a pair is always ↑↓.	5
12	Identify the element described by this configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$ Give its symbol, name and atomic number.	Ar, Argon, Z = 18	Shorthand: [Ne] $3s^2\ 3p^6$. Shells: 2, 8, 8.	5

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